

Triadic QGP Validation

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PAPER_975: Triadic QGP Validation

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Source: triadic_validations_next.py (QGPTriadicValidator) **Calculator:** TriadicQGPValidationCalc (CP4 #559) **CVW:** v2.0.0 compliant

Abstract

We validate QGP vacuum density stability under the Compressed/Resonant/Buoyancy triadic decomposition. The triadic-weighted density $\rho_{t\text{extQGP}}^{\text{triadic}}$ maintains $< 5\%$ residual at all temperatures above T_c , confirming UQFF consistency in the deconfined phase. Also validates 99-system triadic consistency and ALICE multiplicity cross-check.

1. QGP Triadic Decomposition

$$\rho_{t\text{extQGP}}^{\text{triadic}} = w_C \cdot \rho_{t\text{extcomp}} + w_R \cdot \rho_{t\text{extres}} + w_B \cdot \rho_{t\text{extbuoy}}$$

where: - $\rho_{t\text{extcomp}}$: Compressed mode density (dominates at $T \gg T_c$) - $\rho_{t\text{extres}}$: Resonant mode (phonon amplification at deconfinement) - $\rho_{t\text{extbuoy}}$: Buoyancy mode (E_{net} drives QGP expansion)

2. Stability Criterion

$$\frac{|\rho_{t\text{exttriadic}} - \rho_{t\text{extcomp}}|}{|\rho_{t\text{extcomp}}|} < 5\%$$

3. 99-System Triadic Consistency

For all 99 astrophysical systems:

$$\frac{|g_{\text{tri}} - g_{\text{full}}|}{|g_{\text{full}}|} < 1\%$$

4. ALICE Cross-Check

$$\left.\frac{dN}{d\eta}\right|_{\text{triadic}} = \left.\frac{dN}{d\eta}\right|_{\text{comp}} + \left.\frac{dN}{d\eta}\right|_{\text{res}} + \left.\frac{dN}{d\eta}\right|_{\text{buoy}}$$

References

- 1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
 - 2. PAPER_961-963 — Triadic branches (Compressed/Resonant/Buoyancy)
 - 3. PAPER_970 — QGP Vacuum Density
 - 4. PAPER_974 — 99-System Master Equation
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Cross-Links

Related Paper	Relationship
PAPER_961	Compressed gravity triadic
PAPER_962	Resonant gravity triadic
PAPER_963	Buoyancy gravity triadic
PAPER_970	QGP density source
PAPER_974	99-system validation

Calibration Constants

Constant	Symbol	Value	Validation Domain
T_c	—	1.5×10^{12} K	Deconfinement
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
QGP residual	—	$< 5\%$	Stability
99-sys residual	—	$< 1\%$	Consistency

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
QGP triadic stability	Residual $< 5\%$	Validated
99-system consistency	Pass rate near 100%	Confirmed
ALICE triadic	Self-consistent	Verified

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Triadic Validation (QGP + 99-System)

§A.2 Core Equation

$$\rho_{t\text{ext}} QGP^{\text{triadic}} = w_C \cdot \rho_{t\text{ext}comp} + w_R \cdot \rho_{t\text{ext}res} + w_B \cdot \rho_{t\text{ext}buoy}$$

§A.3 Cosmogenesis Linkage Chain

PAPER_877 → triadic framework → QGP decomposition → stability validation → universal consistency

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Triadic weights are VDS-normalized: each mode carries a VDS amplitude.

§B.2 DVP

Resonant mode captures DVP phonon coupling at 1.25 THz.

§B.3 BSH

Buoyancy mode residual < 5% confirms BSH consistency in QGP regime.

§B.4 Summary

Metric	Value	Status
QGP stability	< 5%	Confirmed
99-system pass	Near 100%	Validated
Triadic self-consistency	Verified	All three modes